

An autonomous laboratory for the accelerated synthesis of inorganic materials

<https://doi.org/10.1038/s41586-023-06734-w>

Received: 16 May 2023

Accepted: 10 October 2023

Published online: 29 November 2023

Open access

 Check for updates

Nathan J. Szymanski^{1,2,5}, Bernardus Rendy^{1,2,5}, Yuxing Fei^{1,2,5}, Rishi E. Kumar^{3,5}, Tanjin He^{1,2}, David Milsted², Matthew J. McDermott^{1,2}, Max Gallant^{1,2}, Ekin Dogus Cubuk⁴, Amil Merchant⁴, Haegyeom Kim², Anubhav Jain³, Christopher J. Bartel², Kristin Persson^{1,2}, Yan Zeng²✉ & Gerbrand Ceder^{1,2}✉

To close the gap between the rates of computational screening and experimental realization of novel materials^{1,2}, we introduce the A-Lab, an autonomous laboratory for the solid-state synthesis of inorganic powders. This platform uses computations, historical data from the literature, machine learning (ML) and active learning to plan and interpret the outcomes of experiments performed using robotics. Over 17 days of continuous operation, the A-Lab realized 36 compounds from a set of 57 targets including a variety of oxides and phosphates that were identified using large-scale *ab initio* phase-stability data from the Materials Project and Google DeepMind. Synthesis recipes were proposed by natural-language models trained on the literature and optimized using an active-learning approach grounded in thermodynamics. Analysis of the failed syntheses provides direct and actionable suggestions to improve current techniques for materials screening and synthesis design. The high success rate demonstrates the effectiveness of artificial-intelligence-driven platforms for autonomous materials synthesis and motivates further integration of computations, historical knowledge and robotics.

Although promising new materials can be identified at scale using high-throughput computations, their experimental realization is often challenging and time-consuming. Accelerating the experimental segment of materials discovery requires not only automation but autonomy—the ability of an experimental agent to interpret data and make decisions based on it. Pioneering efforts have demonstrated autonomy in several aspects of materials research, including robotic and Bayesian-driven optimization of carbon nanotube yield^{3,4}, photovoltaic performance⁵ and photocatalysis activity⁶. In contrast to conventional ML algorithms used for optimization, human researchers benefit from a wealth of background knowledge that informs their decision-making, and it is increasingly recognized^{7–9} that autonomy will require a fusion of encoded domain knowledge, access to diverse data sources and active learning.

Here we present the A-Lab, an autonomous laboratory that integrates robotics with the use of *ab initio* databases, ML-driven data interpretation, synthesis heuristics learned from text-mined literature data and active learning to optimize the synthesis of inorganic materials in powder form. Although autonomous workflows based on liquid handling have been demonstrated in organic chemistry^{10–13}, the A-Lab addresses the unique challenges of handling and characterizing solid inorganic powders. These often require milling to ensure good reactivity between precursors, which can have a wide range of physical properties related to differences in their density, flow behaviour, particle size, hardness and compressibility. The use of solid powders is well suited for manufacturing and technological scaleup, and the approach

of the A-Lab to synthesis produces multigram sample quantities that facilitate device-level testing.

Given a set of air-stable target materials (that is, desired synthesis products whose yield we aim to maximize) screened using the Materials Project¹⁴, the A-Lab generates synthesis recipes using ML models trained on historical data from the literature and then performs these recipes with robotics. The synthesis products are characterized by X-ray diffraction (XRD), with two ML models working together to analyse their patterns. When synthesis recipes fail to produce a high target yield, active learning closes the loop by proposing improved follow-up recipes. Over 17 days of operation, the A-Lab successfully synthesized 36 of 57 target materials that span 33 elements and 40 structural prototypes (Supplementary Fig. 1 and Supplementary Table 1). Inspection of the 17 unobtained targets revealed synthetic as well as computational failure modes, several of which could be overcome through minor adjustments to the lab's decision-making. With its high success rate in validating predicted materials, the A-Lab showcases the collective power of *ab initio* computations, ML algorithms, accumulated historical knowledge and automation in experimental research.

Autonomous materials-synthesis platform

The materials synthesis and characterization procedures used in the A-Lab are shown in Fig. 1. All target materials considered in this work are new to the lab, that is, not present in the training data for the algorithms it uses to propose synthesis recipes (Methods). The experiments

¹Department of Materials Science and Engineering, University of California, Berkeley, Berkeley, CA, USA. ²Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ³Energy Technologies Area, Lawrence Berkeley National Laboratory, Berkeley, CA, USA. ⁴Google DeepMind, London, UK. ⁵These authors contributed equally: Nathan J. Szymanski, Bernardus Rendy, Yuxing Fei, Rishi E. Kumar. ✉e-mail: yanzeng@lbl.gov; gceder@berkeley.edu

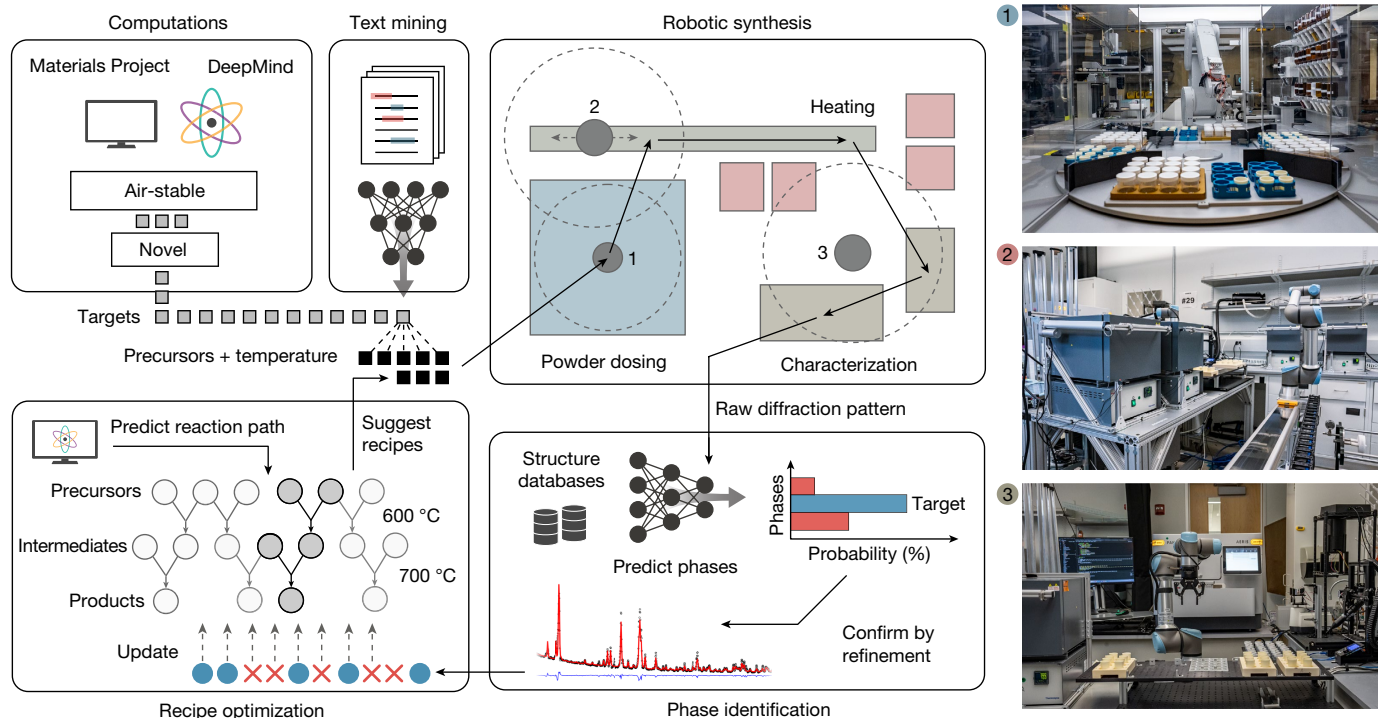


Fig. 1 | Autonomous materials synthesis with the A-Lab. Air-stable targets are identified using DFT-calculated convex hulls consisting of ordered ground states from the Materials Project and Google DeepMind. Synthesis recipes for each target are proposed using ML models trained on synthesis data from the literature. These recipes are tested using a robotic laboratory that automates (1) powder dosing, (2) sample heating and (3) product characterization with XRD. All sample transfer between these stations is performed using robotic arms,

forming a fully automated sequence from chemical input to characterization. Phase purity is assessed from XRD, which is analysed by ML models trained on structures from the Materials Project and the ICSD, and confirmed with automated Rietveld refinement. In cases in which high (>50%) target yield is not obtained, new synthesis recipes are proposed by an active-learning algorithm that identifies reaction pathways with maximal driving force to form the target.

reported in this study represent the first attempts by the A-Lab to synthesize any of these targets. Each target is predicted to be on or very near (<10 meV per atom) the convex hull formed by stable phases taken from the Materials Project¹⁴ and cross-referenced with an analogous database from Google DeepMind. Because the A-Lab handles samples in open air, we only considered targets that are predicted not to react with O₂, CO₂ and H₂O (Methods). Like all databases comprised of atomistic calculations^{15–17}, the Materials Project represents its compounds as ordered structures, even in cases where these materials may form in experiments with partial disorder. This approach is commonly used in theory-driven materials screening efforts as ab-initio computations cannot represent partial atoms^{18,19}. As such, we consider a synthesis procedure successful when it forms either an ordered or partially disordered version of its target material^{20,21}.

For each compound proposed to the A-Lab, up to five initial synthesis recipes are generated by a ML model that has learned to assess target ‘similarity’ through natural-language processing of a large database of syntheses extracted from the literature²², mimicking the approach of a human to base an initial synthesis attempt on analogy to known related materials. A synthesis temperature is then proposed by a second ML model trained on heating data from the literature²³ (Methods). If these literature-inspired recipes fail to produce >50% yield for their desired targets, the A-Lab continues to experiment using Autonomous Reaction Route Optimization with Solid-State Synthesis (ARROWS³), an active-learning algorithm that integrates ab initio computed reaction energies with observed synthesis outcomes to predict solid-state reaction pathways²⁴. Experiments are performed under the guidance of this algorithm until the target is obtained as the majority phase or all synthesis recipes available to the A-Lab are exhausted.

The A-Lab carries out experiments using three integrated stations for sample preparation, heating and characterization, with robotic arms

transferring samples and labware between them (Fig. 1 and Extended Data Figs. 1 and 2). The first station dispenses and mixes precursor powders before transferring them into alumina crucibles. A robotic arm from the second station loads these crucibles into one of four available box furnaces to be heated (Methods). After allowing the samples to cool, another robotic arm transfers them to the third station, where they are ground into a fine powder and measured by XRD. The operations of the lab are controlled through an application programming interface, which enables on-the-fly job submission from human researchers or decision-making agents (Extended Data Fig. 3).

The phase and weight fractions of the synthesis products are extracted from their XRD patterns by probabilistic ML models trained on experimental structures from the Inorganic Crystal Structure Database (ICSD) following the methodology outlined in previous work^{25,26}. Because the A-Lab is given no prior experimental reports for the target materials considered in this work, their diffraction patterns are simulated from computed structures available in the Materials Project and corrected to reduce density functional theory (DFT) errors (Supplementary Note 1). For each sample, the phases identified by ML are confirmed with automated Rietveld refinement (Methods and Supplementary Note 2) and the resulting weight fractions are reported to the management server of the A-Lab to inform subsequent experimental iterations, if necessary, in search of an optimal recipe with high target yield.

Experimental synthesis outcomes

Using the described workflow, the A-Lab synthesized 36 of the 57 target compounds over 17 days of continuous experimentation, representing a 63% success rate. To confirm whether the A-Lab successfully made these targets, all diffraction patterns were later manually refined and

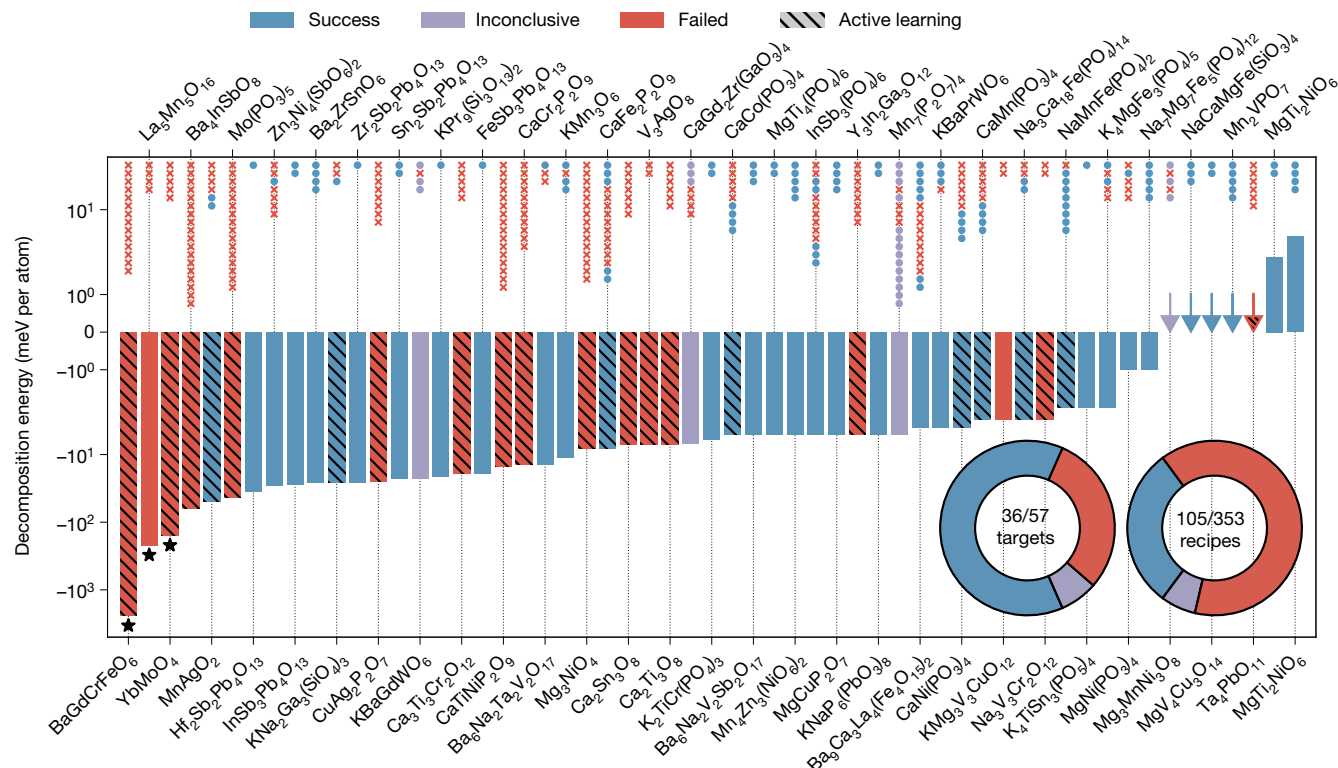


Fig. 2 | Outcomes from targeted syntheses of 57 inorganic materials. Results summary from the synthesis efforts targeting 57 compounds, plotted against their decomposition energies (log scale). Arrows indicate values near zero. A total of 36 targets were successfully synthesized (blue bars), four were inconclusive from XRD (purple bars), and the remaining 17 were not obtained by the A-Lab (red bars). Targets optimized in the active-learning stage of the A-Lab are marked by diagonal lines; all other targets were only attempted using

recipes proposed by ML algorithms trained on literature data. The scatter points above each bar represent the outcomes of attempted recipes for each target, ordered from top to bottom in the sequence in which they were performed. The inset pie charts show the fraction of successful targets (left) and recipes (right). Analyses performed after the fact suggest that the calculated decomposition energies for three targets, marked with stars, may be suspect owing to computational errors (see text).

compared with the A-Lab's autonomously made refinements and conclusions (Supplementary Data). Our manual analysis confirms that in each of the 36 successes, a satisfactory fit to the experimental diffraction pattern can be obtained only when the target is included in refinement. This does not necessarily imply the target was made with high purity, as many of the samples contain prominent byproducts that likely result from incomplete reactions. The A-Lab used a short hold time of 4 h during its syntheses, which could be extended and combined with intermittent regrinding to improve the phase purity. We also note that while the automated XRD interpretation may not identify multi-phase mixtures conclusively, it is generally reliable in identifying the majority phase, which is then used to guide synthesis optimization. We show in the next section that this success rate could be improved to 67% with only minor modifications to the lab's decision-making algorithm, and further to 70% if the computational techniques were also improved. Additionally, there were four materials (totaling 40/57 successes) whose presence in the synthesis products was originally reported by the A-Lab, but manual reexamination deems inconclusive from XRD alone due to ambiguity between the target's diffraction pattern and other known phases.

The high success rate demonstrates that comprehensive *ab initio* calculations can be used to effectively identify synthesizable materials. The outcome for all 57 compounds is plotted in Fig. 2 against their decomposition energies (on a log scale), a common thermodynamic metric that describes the driving force to form a compound from its neighbours on the phase diagram²⁷ (Supplementary Fig. 2). A negative (positive) decomposition energy indicates that a material is stable (metastable) at 0 K. Of the targets considered in this work, 50 are predicted to be stable, whereas the remaining seven are metastable but lie near

the convex hull. Over the range of decomposition energies considered, we do not observe a clear correlation between decomposition energy and whether a material was successfully synthesized.

In total, 30 of the 36 materials synthesized by the A-Lab were obtained using recipes proposed by ML models trained on synthesis data from the literature (Supplementary Note 3). These literature-inspired recipes were more likely to succeed when the reference materials are highly similar to our targets (Supplementary Fig. 3), confirming that target 'similarity' is a useful metric to select effective precursors²⁸. At the same time, precursor selection remains a highly nontrivial task, even for thermodynamically stable materials. Despite 63% of targets eventually being obtained, only 30% of the 353 synthesis recipes tested by the A-Lab produced their targets. This finding echoes previous work that has established the strong influence of precursor selection on the synthesis path, ultimately deciding whether it forms the target or becomes trapped in a metastable state^{29–32}.

The active-learning cycle of the A-Lab²⁴ identified synthesis routes with improved yield for nine targets, of which six had zero yield from the initial literature-inspired recipes. Targets optimized with active learning are indicated by the bars containing diagonal lines in Fig. 2. In this framework, improved synthesis routes are designed using two hypotheses: (1) solid-state reactions tend to occur between two phases at a time (that is, pairwise)^{33–35} and (2) intermediate phases that leave only a small driving force to form the target material should be avoided, as they often require long reaction time and high temperature^{29,30,36}.

The A-Lab continuously builds a database of pairwise reactions observed in its experiments—88 unique pairwise reactions (Supplementary Table 2) were identified from the synthesis experiments performed in this work. This database allows the products of some recipes to be

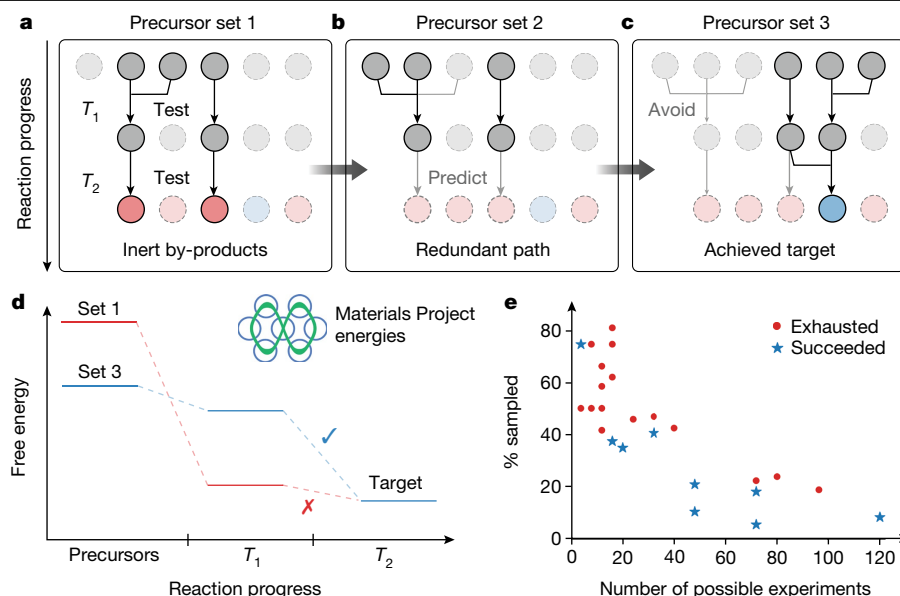


Fig. 3 | Active learning with pairwise reaction analysis. **a**, From a failed synthesis attempt, the A-Lab determines which pairwise reactions occurred. **b**, New precursors are recommended by substituting at least one precursor involved in the unfavourable pairwise reaction. In cases in which the new precursor set leads to identical intermediates as a previously tested set, it is not explored at any higher temperatures. **c**, The successful precursor set avoids all the unfavourable pairwise reactions. **d**, The free energy at each step

in the reaction pathway, calculated using data from the Materials Project, which shows that the successful pathway maintains a large driving force at the target-forming step. **e**, Following this approach, the scatter plot shows the number of experiments required to exhaust all unique reaction paths for each target (red) or to identify an optimal path with high yield (blue), plotted with respect to the total size of each experimental search space.

inferred, precluding their testing; a recipe that yields an observed set of intermediates (already present in the lab's database) need not be pursued at higher temperatures, as the remaining reaction pathway is already known (Fig. 3a,b). This can reduce the search space of possible synthesis recipes by up to 80% when many precursor sets react to form the same intermediates (Fig. 3e and Supplementary Notes 4 and 5). Furthermore, knowledge of reaction pathways can be used to give priority to intermediates with a large driving force to form the target, computed using formation energies available in the Materials Project (Fig. 3c,d). For example, the synthesis of $\text{CaFe}_2\text{P}_2\text{O}_9$ was optimized by avoiding the formation of FePO_4 and $\text{Ca}_3(\text{PO}_4)_2$, which have a small driving force (8 meV per atom) to form the target. This led to the identification of an alternative synthesis route that forms $\text{CaFe}_3\text{P}_3\text{O}_{13}$ as an intermediate, from which there remains a much larger driving force (77 meV per atom) to react with CaO and form $\text{CaFe}_2\text{P}_2\text{O}_9$, causing an approximately 70% increase in the yield of the target (Supplementary Note 6).

Barriers to synthesis

Seventeen of the 57 targets evaluated by the A-Lab were not obtained even after its active-learning cycle. We identify slow reaction kinetics, precursor volatility, amorphization and computational inaccuracy as four broad categories of 'failure modes' that prevented the synthesis of these targets. The prevalence of each failure mode is shown in Fig. 4, accompanied by their affected targets.

Sluggish reaction kinetics hindered 11 of the 17 failed targets, each containing reaction steps with low driving forces (<50 meV per atom; Supplementary Fig. 4). In principle, these targets can be made accessible by using a higher synthesis temperature, longer heating time, improved precursor mixing or intermittent regrinding—standard procedures that are at present outside the domain of the A-Lab's active-learning algorithm. As such, we manually reground the original synthesis products generated by the A-Lab and heated them to higher temperatures, which led to the successful formation of two further targets, $\text{Y}_3\text{Ga}_3\text{In}_2\text{O}_{12}$ and Mg_3NiO_4 , bringing our total success rate to 67% (Supplementary Note 7). One could also use more reactive precursors

to provide a greater driving force to form the target, although our experiments were constrained to air-stable binary precursors that sometimes restricted the A-Lab's choice of synthesis routes to those forming highly stable intermediates. System modifications to enable multistep heating, intermediate regrinding and expanded precursor selection should improve the ability of the lab to adapt and overcome failed synthesis attempts.

Precursor volatility disrupted all synthesis experiments targeting $\text{CaCr}_2\text{P}_2\text{O}_9$, causing a change in the net stoichiometry of its samples (Supplementary Note 8). This can be attributed to the use of ammonium phosphate precursors, $\text{NH}_4\text{H}_2\text{PO}_4$ and $(\text{NH}_4)_2\text{HPO}_4$, which proceed through a series of decomposition reactions and ultimately evaporate above 450 °C (ref. ³⁷). Still, recipes based on these precursors can succeed if the ammonium phosphate reacts with another precursor before its evaporation temperature, effectively locking the phosphate ions in the solid state. For example, volatility does not seem to be an issue for the Mn-containing phosphates targeted in this work, as each Mn oxides precursor reacts with the ammonium phosphates at low temperature (<500 °C) to form $\text{Mn}_2(\text{PO}_4)_3$ as an intermediate. This precursor behaviour can, in principle, be learned when sufficient pairwise reaction data have been collected, after which the A-Lab may favour the selection of precursors that trap in phosphate ions at low temperature and therefore preclude unwanted volatility.

Melting of samples at high temperature inhibited the crystallization of one target, $\text{Mo}(\text{PO}_3)_5$, whose synthesis attempts produced amorphous samples (Supplementary Fig. 5). Although the use of a molten flux can sometimes improve reaction kinetics³⁸, the formation of an amorphous state that is low in energy may reduce the driving force for crystallization. Indeed, using the workflow outlined in ref. ³⁹, we identified amorphous configurations of $\text{Mo}(\text{PO}_3)_5$ with energies as low as 61 meV per atom above the crystalline ground state, a finding that is consistent with the widely reported glass-forming ability of phosphate-rich compounds^{40,41}.

Some failure modes result from inaccuracies in the computed stability of the target and therefore cannot be addressed by modifications to the experimental procedures. Fundamental electronic-structure

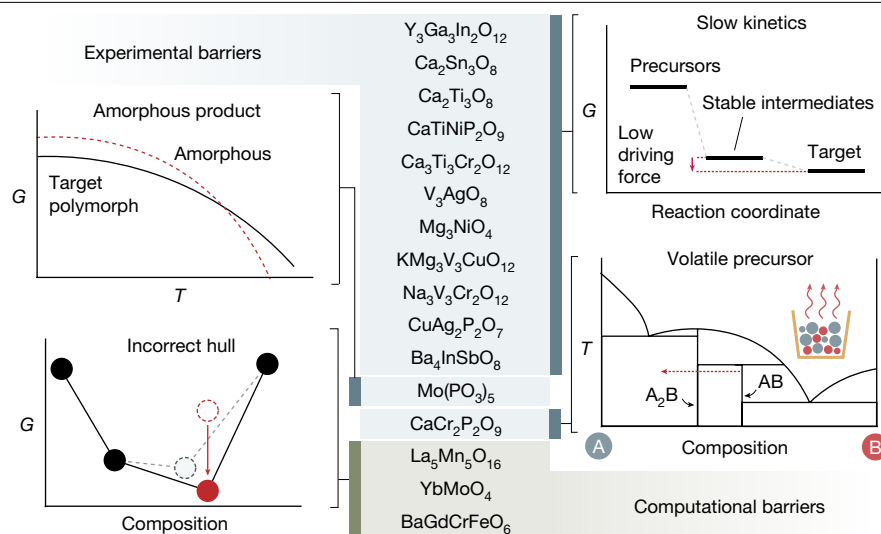


Fig. 4 | Barriers to the synthesis of materials predicted to be stable. The 17 target materials that could not be synthesized by the A-Lab, for which each is categorized by the feature that complicates its synthesis. One target ($\text{Ta}_4\text{Pb}_{11}$) is excluded from this list, as it is metastable and therefore was predictably unobtained in favour of its stable competitors. The challenges in synthesizing the remaining 16 stable targets fall under two categories: experimental barriers

(blue, 13 targets) and computational barriers (green, three targets). We distinguish these barriers as four distinct failure modes: slow reaction kinetics, precursor volatility, product amorphization and limitations associated with DFT calculations performed at 0 K. A schematic explanation for each failure mode is provided.

challenges are probably affecting $\text{La}_5\text{Mn}_5\text{O}_{16}$, as all the attempts to synthesize this phase instead yielded LaMnO_3 , which DFT unexpectedly predicts to be highly unstable (120 meV per atom above the hull), even though it is widely reported in the literature to be experimentally accessible⁴². If the energy of LaMnO_3 were lowered, consistent with its experimental stability, $\text{La}_5\text{Mn}_5\text{O}_{16}$ would be destabilized (above the hull). Errors in the computed energy of LaMnO_3 may arise from its strong Jahn–Teller activity⁴³, compositional off-stoichiometry⁴⁴ or the presence of f-states in La—all of which present challenges to conventional DFT. Problems with YbMoO_4 were found to be because of a poor pseudopotential choice in the Materials Project that destabilizes the well-known oxide, Yb_2O_3 , and it is likely that, in more accurate calculations, YbMoO_4 is not stable. A similar lanthanide-related electronic-structure problem may also be responsible for the failure to synthesize BaGdCrFeO_6 . These examples demonstrate the ability of the A-Lab to provide important feedback to high-throughput computed datasets. With improved calculations that exclude the computationally problematic compounds in this work, our total success rate would increase to 70% (38/54 targets).

Outlook

In 17 days of closed-loop operation, the A-Lab performed 353 experiments and successfully realized 36 of 57 inorganic crystalline solids with diverse structures and chemistries. This high success rate (63%) was achieved by integrating robotics with: (1) DFT-computed data to survey the energetic landscape of precursors, reaction intermediates and final products; (2) heuristic suggestions for synthesis procedures obtained from ML models trained on text-mined synthesis data; (3) ML interpretation of experimental data; and (4) an active-learning algorithm that improves on failed synthesis procedures. The study also revealed several opportunities to enhance the lab's active-learning algorithm by addressing failures caused by slow reaction kinetics, which would enable an improved success rate of 67% with in-line solutions.

Our paper demonstrates that autonomous research agents can markedly accelerate the pace of materials research. Researchers initialized the A-Lab by proposing 57 target materials, which were successfully realized at a rate of >2 additional materials per day with minimal human intervention. Such rapid and successful experimentation point to a vast

landscape of opportunities in materials synthesis and development. Although this work focused on a limited subset of all possible synthesis targets, many new candidates await evaluation. As the breadth of ab initio computations continues to grow, so will this list of materials.

While the characterization of multi-phase samples remains challenging, decision making in the A-Lab is supported by its robust identification of the majority phase in each sample, combined with the use of computed phase stability data. Nonetheless, there remains ample room for improvement in the automated interpretation of materials characterization data. Algorithms that can more effectively account for impurity phases in XRD patterns, augmented with additional characterization techniques such as electron microscopy and mass spectrometry, would benefit the next generation of autonomous research platforms.

Advances in simulations, ML and robotics have intersected to enable 'expert systems' that show autonomy as an emergent quality by the sum of its automated components. The A-Lab demonstrates this by combining modern theory-driven and data-driven ML techniques with a modular workflow that can synthesize materials with minimal human input. Lessons learned from continuing experiments can inform both the system itself and the greater community through systematic data generation and collection. The systematic nature of the A-Lab provides a unique opportunity to answer fundamental questions about the factors that govern the synthesizability of materials, serving as an experimental oracle to validate predictions made on the basis of data-rich resources such as the Materials Project. In future iterations of the platform, such an oracle may be expanded to investigate factors beyond synthesizability, including microstructure and device performance. Although our current success rate is high, the remaining discrepancies between current predictions and their experimental outcomes is a crucial signal required to improve our understanding of materials synthesis.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06734-w>.